

The analysis of *Acrocephalus paludicola* samples followed the general bioinformatic pipeline with the following deviations:

1. Software. The analysis was conducted on a HPC system which required working with containers. For this reason, singularity containers were used, with versions corresponding to the general pipeline:

bwa.0.7.19.sif https://depot.galaxyproject.org/singularity/bwa:0.7.19--h577a1d6_1

samtools.1.21.sif https://depot.galaxyproject.org/singularity/samtools:1.21--h96c455f_1

gatk4_4.6.2.0.sif docker://broadinstitute/gatk:4.6.2.0

plink2_00a3.3.sif docker://quay.io/biocontainers/plink2:2.00a3.3--hb2a7ceb_0

vcftools.0.1.17.sif docker://quay.io/biocontainers/vcftools:0.1.17--pl5321h077b44d_0

bcftools.1.21.sif docker://quay.io/biocontainers/bcftools:1.21--h3a4d415_1

repeatmasker.4.1.9.sif docker://quay.io/biocontainers/repeatmasker:4.1.9--pl5321hdfd78af_0

fastqc.0.12.1.sif docker://quay.io/biocontainers/fastqc:0.12.1--hdfd78af_0

multiqc.1.29.sif docker://quay.io/biocontainers/multiqc:1.29--pyhdfd78af_0

fastp v1.0.1 was installed from binaries: <http://opengene.org/fastp/fastp>

2. Pipeline.

- **Repeatmasker.** Option -species passeriformes was used, from the famdb library https://www.dfam.org/releases/current/families/FamDB/dfam39_full.2.h5.gz
- **Alignment.** Alignment, conversion to .bam, adding read groups, merging files from different lanes by sample, sorting and indexing were done as follows:
 - bwa mem -M -t 8 \${asm}.soft.fa \$r1 \$r2 > \${sampleFile}.\${asm}.sam
 - samtools view -@ 8 -b \${sample}.\${asm}.sam -o \${sampleFile}.\${asm}.bam
 - samtools addreplacerg -r ID:\${SAMPLE}_\${LANE} -r SM:\${SAMPLE} -r PL:ILLUMINA -r LB:\${SAMPLE} -r PU:\${FLOWCELL}.\${LANE} -o \${sampleFile}.\${asm}_tagged.bam \${sampleFile}.\${asm}.bam
 - samtools merge -@ 8 -f -o \${sample}.\${asm}_merged.bam \${sampleFile}*\${asm}_tagged.bam
 - samtools sort -@ 8 \${sample}.\${asm}_merged.bam -o \${sample}.\${asm}_sorted.bam
 - samtools index -@ 8 \${sample}.\${asm}_sorted.bam
- **Duplicate marking & indexing.** A reordering step was added.
 - gatk --java-options "-Xmx25g -XX:ParallelGCThreads=8" MarkDuplicates -I \${sample}.\${asm}_sorted.bam -O \${sample}_\${asm}_dedup.bam -M \${sample}.\${asm}_dup.txt
 - java -jar /usr/picard/picard.jar ReorderSam -I \${sample}_\${asm}_dedup.bam -O \${sample}_\${asm}_reordered.bam -R \${asm}.masked.fa -SEQUENCE_DICTIONARY \${asm}.masked.dict
 - samtools index \${sample}_\${asm}_reordered.bam

- **Haplotype calling.** Done on reordered samples, one job per sample.
 - `gatk --java-options "-Xmx50g -XX:ParallelGCThreads=16" HaplotypeCaller -R ${asm}.masked.fa -I ${sample}_${asm}_reordered.bam -ERC GVCF -O ${sample}.${asm}.g.vcf.gz`
- **Combining sample-level GVCFs.** Done in batches of 2-3 samples for each assembly. The batches were then combined. For example:
 - `gatk CombineGVCFs --java-options "-Xmx70g -XX:ParallelGCThreads=16" -R ${asm}.masked.fa --verbosity DEBUG --QUIET false --variant sample1.${asm}.g.vcf.gz --variant sample2.${asm}.g.vcf.gz --variant sample3.${asm}.g.vcf.gz -O cohort1.sample1.sample2.sample3.${asm}.g.vcf.gz`
 - `gatk CombineGVCFs --java-options "-Xmx70g -XX:ParallelGCThreads=16" -R ${asm}.masked.fa --verbosity DEBUG --QUIET false --variant sample4.${asm}.g.vcf.gz --variant sample5.${asm}.g.vcf.gz -O cohort2.sample4.sample5.${asm}.g.vcf.gz`
 - `gatk CombineGVCFs --java-options "-Xmx70g -XX:ParallelGCThreads=16" -R ${asm}.masked.fa --verbosity DEBUG --QUIET false --variant cohort1.sample1.sample2.sample3.${asm}.g.vcf.gz --variant cohort2.sample4.sample5.${asm}.g.vcf.gz -O combined_all_cohorts.${asm}.g.vcf.gz`
- **SNP filtering.** The bcftools step was slightly modified:
 - `bcftools view -f PASS -m2 -M2 -v snps -Oz -o cohort.${asm}.flt.bcftools.vcf.gz combined_all_cohorts.${asm}.flt.vcf.gz`
 - `bcftools index cohort.${asm}.flt.bcftools.vcf.gz`
- **Population genomics analysis: LD pruning and PCA.**
 - `plink2 --vcf cohort.${asm}.final.vcf.gz --indep-pairwise 50 10 0.2 --bad-ld --out pruned_${asm} --allow-extra-chr --set-all-var-ids @:#:\$r:\$a`
 - `vcftools --vcf cohort.${asm}.final.vcf.gz --set-all-var-ids @:#:\$r:\$a --extract pruned_${asm}.prune.in --read-freq ${asm}.freq.afreq --pca 10 --out pca_${asm} --allow-extra-chr`
- **Diversity statistics.**
 - `Extract SNPs to remove: awk -F ' ' '{print $1"\t"$2}' pruned_${asm}.prune.out > exclude_snps_${asm}.txt`
 - `vcftools --gzvcf cohort.${asm}.final.vcf.gz --exclude-positions exclude_snps_${asm}.txt --site-pi --out pi_${asm}`
 - `vcftools --gzvcf cohort.${asm}.final.vcf.gz --exclude-positions exclude_snps_${asm}.txt --het --out het_${asm}`
 - `vcftools --gzvcf cohort.${asm}.final.vcf.gz --exclude-positions exclude_snps_${asm}.txt --weir-fst-pop pop1.txt --weir-fst-pop pop2.txt --out fst_${asm}`